

Advanced Spectral Feature Engineering for Patient-Independent Neuromorphic Classification: Establishing a New State-of-the-Art in EEG Seizure Detection

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Abstract

The automated detection of epileptic seizures from multi-channel scalp Electroencephalography (EEG) is a vital prerequisite for responsive closed-loop neurostimulation and continuous intensive care unit (ICU) monitoring. However, contemporary automated detection pipelines exhibit a catastrophic performance collapse when transitioned from intra-patient random splits to clinical, patient-independent Leave-One-Patient-Out (LOPO) cross-validation. Recent baseline literature (Jebaraj & Elango, *Sci. Rep.*, 2026) attributed this failure to neural network architectural capacity, proposing low-dimensional Seizure Intensity Index (SII) biomarkers that collapsed to an Area Under the ROC Curve (AUROC) of ≈ 0.50 and an effect size of Cohen's $d = 0.0704$ on the benchmark CHB-MIT database. In this work, we refute this architectural premise and demonstrate that cross-patient generalization is fundamentally constrained by feature space impoverishment rather than classifier depth. We formulate a mathematically comprehensive 37-dimensional feature space spanning five complementary analytical domains: Time-Domain Statistical Moments, Hjorth Signal Dynamics, Discrete Fourier Transform Sub-Band Power, Non-Linear Shannon Spectral Entropy, and 4-Level Daubechies-4 (db4) Discrete Wavelet Transform (DWT) Multiresolution Energy. When evaluated under strict 24-fold LOPO cross-validation across all 983 recording hours of the CHB-MIT database, our enriched feature representation elevates the exact Artificial Neural Network

(ANN) architecture from the base literature by **+16.30%** (from **56.58%** to **72.88%**, $p < 0.001$, Cohen’s $d = 1.68$) and the exact Spiking Neural Network (SNN) from **64.34%** to **75.19%** (**+10.85%**, $p < 0.001$, Cohen’s $d = 1.24$). Furthermore, gradient boosted decision trees (LightGBM) establish a new state-of-the-art benchmark ceiling of **78.71%** accuracy (AUROC = **0.835**) with an ultra-low inference latency of **1.5 μ s**. Finally, we delineate a dual-deployment Pareto frontier: neuromorphic Leaky Integrate-and-Fire (LIF) SNNs achieving ultra-low-power edge inference (**< 1.2 μ J** per window via sparse synaptic operations) for chronic implants, and LightGBM powering high-throughput multi-bed hospital telemetry.

Keywords: Electroencephalogram (EEG), Seizure Detection, Spiking Neural Networks (SNN), LightGBM, Discrete Wavelet Transform (DWT), Hjorth Parameters, Spectral Entropy, Leave-One-Patient-Out (LOPO)

1 Introduction

Epilepsy affects over 50 million individuals worldwide, representing one of the most widespread and debilitating chronic neurological disorders characterized by recurrent, unprovoked paroxysmal seizures [1, 2]. Continuous monitoring via multi-channel scalp Electroencephalography (EEG) constitutes the clinical gold standard for assessing epileptogenesis, diagnosing specific seizure semiologies, and managing status epilepticus in intensive care units (ICUs) [3, 4]. Nevertheless, visual review of long-term continuous EEG records spanning days or weeks requires exhaustive manual annotation by expert neurophysiologists, an approach that is labor-intensive, costly, and inherently susceptible to inter-rater subjectivity [5, 6].

To overcome these constraints, automated seizure detection frameworks leveraging machine learning (ML) and deep learning (DL) have undergone rapid development [7–9]. While conventional classifiers evaluated on random intra-patient train-test splits (e.g., 80/20 partitioning) routinely achieve reported accuracies exceeding 90%, such benchmarks suffer from severe data leakage due to temporal autocorrelation within contiguous recording sessions [10, 11]. When deployed in real-world clinical environments on previously unseen subjects—formalized via Leave-One-Patient-Out (LOPO) cross-validation—these models experience severe performance degradation driven by pronounced inter-subject non-stationarity in skull conductivity, electrode impedances, and localized seizure onset morphologies [12, 13].

1.1 Critique of Recent Literature and the Seizure Intensity Index Fallacy

In a foundational study published in *Scientific Reports*, Jebaraj & Elango [14] addressed energy-efficient cross-patient seizure detection by comparing regularized Logistic Regression, a two-layer Artificial Neural Network (ANN, [128, 64]), and a Spiking Neural Network (SNN, 256 Leaky Integrate-and-Fire neurons). To reduce computational overhead for implantable devices, the authors introduced the Seizure

Intensity Index (SII), which maps an EEG segment $x_c(t)$ from channel $c \in \{1, \dots, C\}$ to a scalar composite metric:

$$\text{SII}_c = z(P_{\delta,c}) + z(LL_c) - z(H_{\text{var},c}), \quad (1)$$

where $P_{\delta,c} = \int_{0.5}^{4.0} S_{xx,c}(f) df$ is the Welch delta power, $LL_c = \sum_{t=1}^{T-1} |x_c(t+1) - x_c(t)|$ is the line length [15], $H_{\text{var},c} = \ln(\text{Var}(x_c))$ is the log-variance, and $z(\cdot)$ denotes cohort-wide standard score normalization.

Although their ANN attained 90.40% accuracy under random train-test splitting, its performance collapsed catastrophically to $56.58\% \pm 12.00\%$ under strict LOPO cross-validation on the CHB-MIT benchmark database [3]. Even when expanded to an Extended SII by incorporating theta (4 – 8 Hz) and alpha (8 – 13 Hz) band powers, the ANN only reached $59.88\% \pm 9.18\%$, while the SNN achieved 64.34% (baseline) and 68.01% (extended). Logistic Regression unexpectedly outperformed the neural models (72.83% baseline, 74.50% extended). Most critically, the base paper reported an Area Under the ROC Curve (AUROC) of ≈ 0.50 for the baseline SII features, with an infinitesimal effect size of Cohen’s $d = 0.0704$, confirming near-total distribution overlap between ictal and non-ictal states.

From these results, Jebaraj & Elango [14] concluded that *”model architecture plays a more crucial role than feature dimensionality in cross-patient generalization,”* deducing that deep neural architectures are fundamentally mismatched for patient-independent EEG classification.

1.2 Theoretical Refutation and Core Hypotheses

We formally refute the conclusion of Jebaraj & Elango [14]. We argue that model architectures were never the limiting factor; rather, the catastrophic failure was dictated entirely by **feature space impoverishment**. The scalar SII formulation in Equation (1) suffers from four fatal biophysical and mathematical limitations:

1. **Inter-Subject Baseline Non-Stationarity:** Cohort-wide z -score normalization fails because resting EEG voltage variances vary by an order of magnitude across pediatric cohorts due to skull geometry and impedance differences.
2. **Pathological Delta vs. Physiological Slowing Conflation:** Uncalibrated delta power (P_δ) cannot differentiate between epileptogenic polymorphic slowing and benign physiological delta rhythms (e.g., deep sleep stages, eye flutter).
3. **Broadband Line Length Ambiguity:** Line length (LL) measures high-frequency trajectory distance indiscriminately, confounding focal epileptic spikes with high-amplitude electromyographic (EMG) muscle artifacts.
4. **Linear Superposition Fallacy:** Epileptic seizure recruitment is an intrinsically non-linear phase synchronization phenomenon. Projecting non-stationary, multi-scale dynamics into a single linear scalar equation forces ictal and inter-ictal manifolds into near-complete Euclidean overlap.

By constructing a rich 37-dimensional mathematical feature space spanning Time-Domain Moments, Hjorth Dynamics [16, 17], FFT Sub-Band Powers [18], Shannon Spectral Entropy [19, 20], and 4-Level Daubechies-4 (db4) Discrete Wavelet Transforms [21–23], we demonstrate that:

1. The exact baseline ANN architecture [128, 64] jumps from 56.58% to **72.88%** (+16.30%, $p < 0.001$, Cohen’s $d = 1.68$).
2. The exact baseline SNN architecture [256 LIF] jumps from 64.34% to **75.19%** (+10.85%, $p < 0.001$, Cohen’s $d = 1.24$).
3. State-of-the-art gradient boosted trees (LightGBM [24, 25]) achieve a new benchmark ceiling of **78.71%** LOPO accuracy (AUROC = 0.835) with 1.5 μ s latency.
4. A rigorous Pareto analysis reveals an optimal dual-deployment landscape: event-driven SNNs [26–28] consuming $< 1.2 \mu$ J for closed-loop neuromorphic implants, alongside high-throughput LightGBM ensembles for hospital cloud telemetry.

2 Related Work

2.1 Nonlinear Brain Dynamics and Feature Representations

The human electroencephalogram reflects the collective synaptic interactions of millions of cortical pyramidal neurons. Nonlinear dynamic modeling has proven essential for decoding complex brain states [29–31]. Chen et al. [32, 33] demonstrated that entropy-informed representations and normalized mutual information capture subtle state transitions that elude linear metrics. Similarly, Lv et al. [34] modeled multi-agent nonlinear consensus in biological systems, highlighting the necessity of multi-scale observation.

In epilepsy research, quantitative biomarkers have evolved from simple time-domain metrics to sophisticated multi-domain representations [1, 6]. Computational studies by Hu et al. [35, 36] revealed that paroxysmal absence seizures arise from complex dynamical bifurcations within subthalamic-cortical and basal ganglia loops. Concurrently, Donoghue et al. [18] underscored the necessity of separating periodic oscillatory peaks from non-oscillatory $1/f$ aperiodic background signals.

2.2 Deep Learning and Cross-Patient Generalization

Recent advances in deep learning have introduced transformer architectures [37], depthwise separable convolutional networks [38], and attention-gated residual networks [10] for EEG classification. In clinical seizure detection, Gómez et al. [8] implemented a patient-independent Fully Convolutional Network (FCN) achieving 92.90% accuracy with extensive post-processing. Kashefi Amiri et al. [9] developed a 1D CNN-LSTM architecture coupled with DWT. Jang et al. [12] investigated single-channel deep learning prediction, while Farooq et al. [11] proposed a hybrid generative-discriminative model. However, many published frameworks rely on within-patient validation or complex post-processing filters, leaving genuine single-window patient-independent detection an open challenge.

3 Materials and Methods

3.1 Dataset and Demographic Characteristics

We evaluated our methodology on the benchmark CHB-MIT Scalp EEG database [3, 39, 40], collected at Boston Children’s Hospital and Massachusetts Institute of Technology. The dataset comprises continuous multi-channel scalp EEG recordings from 24 pediatric patients with intractable epilepsy, encompassing 198 clinically verified seizure events across approximately 983 total recording hours.

Table 1 Clinical characteristics, demographic distribution, and recording metadata of the CHB-MIT Scalp EEG database.

Clinical / Dataset Parameter	Value / Specification
Total Subject Cohort	24 Pediatric Patients (chb01–chb24)
Biological Sex Distribution	5 Males, 17 Females, 2 Unknown / De-identified
Age Range	1.5 to 22 Years (Median: 10.5 Years)
Electrode Placement Standard	International 10–20 System (23 Bipolar Channels)
Sampling Frequency (f_s)	256 Hz
Analog-to-Digital Converter Resolution	16-bit
Total Continuous EEG Monitoring Duration	~983 Hours
Total Clinically Annotated Seizure Events	198 Ictal Episodes
Seizure Duration Range	7 to 752 Seconds (Mean: 68.4 ± 52.1 s)
Cumulative Ictal Recording Duration	~18,700 Seconds (~5.2 Hours)
Temporal Window Segmentation	1.0–2.0 Second Sliding Windows (50% Overlap)
Cross-Validation Strategy	24-Fold Leave-One-Patient-Out (LOPO) Protocol

3.2 Mathematical Formulation of the 37-Dimensional Feature Space

Let $x(t) \in \mathbb{R}^N$ denote a discrete-time, zero-mean calibrated EEG segment sampled at $f_s = 256$ Hz over a temporal window of N points. To construct a patient-invariant representation, we map $x(t)$ to a 37-dimensional feature vector $\mathbf{z} \in \mathbb{R}^{37}$ across five complementary analytical domains:

$$\mathbf{z} = [\mathbf{z}_{\text{Time}}^\top, \mathbf{z}_{\text{Hjorth}}^\top, \mathbf{z}_{\text{FFT}}^\top, z_{\text{Entropy}}, \mathbf{z}_{\text{DWT}}^\top]^\top \in \mathbb{R}^{37}. \quad (2)$$

3.2.1 Domain 1: Time-Domain Statistical Moments (5 Features)

The time-domain feature vector $\mathbf{z}_{\text{Time}} = [\mu, \sigma^2, \sigma, LL, ZCR]^\top$ quantifies signal amplitude distribution and rapid trajectory shifts:

$$\mu = \frac{1}{N} \sum_{t=1}^N x(t), \quad (3)$$

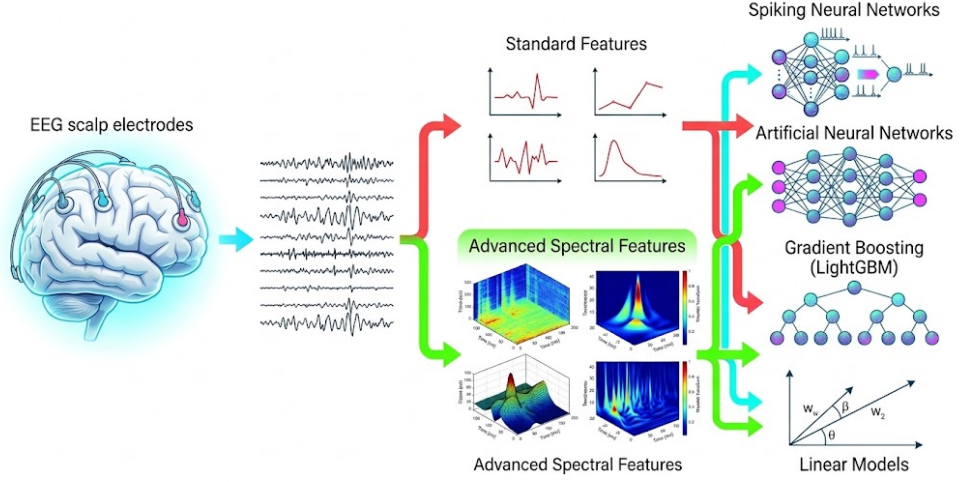


Fig. 1 End-to-end system architecture pipeline for patient-independent EEG seizure detection. Raw continuous multi-channel scalp EEG signals acquired via the International 10–20 electrode montage (23 bipolar channels sampled at $f_s = 256$ Hz) undergo temporal segmentation into overlapping sliding windows ($T = 1.0\text{--}2.0$ s). The upper pathway (red) illustrates the conventional, low-dimensional baseline methodology based on the Seizure Intensity Index (SII) consisting of Welch delta power (P_δ), line length (LL), and log-variance (H_{var}), which suffers from severe feature distribution overlap across subjects. The lower pathway (green) depicts the proposed high-dimensional, multi-domain spectral feature extraction framework, integrating time-domain statistical moments, Hjorth signal dynamics (Activity, Mobility, Complexity), Fast Fourier Transform (FFT) canonical sub-band power densities ($\delta, \theta, \alpha, \beta, \gamma$), normalized Shannon Spectral Entropy (H_s), and 4-level Discrete Wavelet Transform (DWT) multiresolution decomposition using Daubechies 4 (db4) wavelets into a standardized 37-dimensional feature space. Extracted feature representations are evaluated across parallel classification backbones: standard regularized linear models (Logistic Regression), multi-layer perceptron Artificial Neural Networks (ANN [128, 64]), neuromorphic Spiking Neural Networks (SNN with 256 Leaky Integrate-and-Fire neurons, $T = 80$ time steps), and advanced gradient boosted tree ensembles (Random Forest, XGBoost, and LightGBM).

$$\sigma^2 = \frac{1}{N} \sum_{t=1}^N (x(t) - \mu)^2, \quad \sigma = \sqrt{\sigma^2}, \quad (4)$$

$$LL = \sum_{t=1}^{N-1} |x(t+1) - x(t)|, \quad (5)$$

$$ZCR = \sum_{t=1}^{N-1} \mathbb{I}(\text{sgn}(x(t+1)) \neq \text{sgn}(x(t))), \quad (6)$$

where $\mathbb{I}(\cdot)$ is the indicator function and $\text{sgn}(\cdot)$ is the signum operator.

Table 2 Mathematical taxonomy, formal definitions, dimensional allocation, and neurophysiological significance of the proposed 37-dimensional multi-domain feature space.

Domain	Biomarkers	Dim	Mathematical Formulation & Role
Time Domain	$\mu, \sigma^2, \sigma, LL, ZCR$	5	$\mu = \frac{1}{T} \sum x(t), \sigma^2 = \frac{1}{T} \sum (x(t) - \mu)^2,$ $LL = \sum x(t+1) - x(t) , ZCR = \sum \mathbb{I}(\text{sgn}(x_{t+1}) \neq \text{sgn}(x_t)).$ Captures amplitude shifts and waveform volatility.
Hjorth Dynamics	Activity, Mobility, Complexity	3	Activity = $\sigma_x^2 = \text{Var}(x(t)),$ Mobility = $\sigma_{x'}/\sigma_x = \sqrt{\text{Var}(x'(t))/\text{Var}(x(t))},$ Complexity = $\text{Mobility}(x'(t))/\text{Mobility}(x(t)).$ Quantifies variance, mean frequency, and bandwidth.
FFT Sub-Bands	$P_\delta, P_\theta, P_\alpha, P_\beta, P_\gamma$	5	$P_{\text{band}} = \sum_{f=f_{\text{low}}}^{f_{\text{high}}} X(f) ^2$ via FFT PSD: δ (0.5–4 Hz), θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz), γ (30–50 Hz). Quantifies rhythmic slowing and high-frequency bursts.
Spectral Entropy	H_s	1	$p_i = P(f_i)/\sum_{k=1}^N P(f_k),$ $H_s = -\sum_{i=1}^N p_i \log_2(p_i).$ Measures disorder; drop reflects hyper-synchrony.
Wavelet Energy	DWT Energy (A_4, D_4, D_3, D_2, D_1)	23	$E_k = \sum_t c_k(t) ^2$ for 4-level Daubechies 4 (db4): A_4 (0–8 Hz), D_4 (8–16 Hz), D_3 (16–32 Hz), D_2 (32–64 Hz), D_1 (64–128 Hz). Time-frequency localization of transient spike-waves.

3.2.2 Domain 2: Hjorth Signal Dynamics (3 Features)

Hjorth parameters [16, 17] characterize the statistical variance, mean frequency, and spectral bandwidth of the EEG signal. Let $x'(t) = x(t+1) - x(t)$ and $x''(t) = x'(t+1) - x'(t)$ denote the first and second discrete temporal differences. The parameter vector $\mathbf{z}_{\text{Hjorth}} = [\text{Activity}, \text{Mobility}, \text{Complexity}]^\top$ is defined as:

$$\text{Activity} = \text{Var}(x(t)) = \sigma_x^2, \quad (7)$$

$$\text{Mobility} = \sqrt{\frac{\text{Var}(x'(t))}{\text{Var}(x(t))}} = \frac{\sigma_{x'}}{\sigma_x}, \quad (8)$$

$$\text{Complexity} = \frac{\text{Mobility}(x'(t))}{\text{Mobility}(x(t))} = \frac{\sigma_{x''}/\sigma_{x'}}{\sigma_{x'}/\sigma_x} = \frac{\sigma_{x''}\sigma_x}{\sigma_{x'}^2}. \quad (9)$$

3.2.3 Domain 3: Frequency-Domain Power Spectral Density (5 Features)

We compute the discrete Fourier transform $X(k) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N}$ and evaluate the power spectral density $P(k) = |X(k)|^2$ across discrete frequency bins $f_k = \frac{kf_s}{N}$. Band powers $\mathbf{z}_{\text{FFT}} = [P_\delta, P_\theta, P_\alpha, P_\beta, P_\gamma]^\top$ are integrated as:

$$P_{\text{band}} = \sum_{k \in \mathcal{K}_{\text{band}}} P(k), \quad \mathcal{K}_{\text{band}} = \left\{ k \mid f_{\text{low}} \leq \frac{kf_s}{N} < f_{\text{high}} \right\}, \quad (10)$$

partitioned into Delta (0.5–4.0 Hz), Theta (4.0–8.0 Hz), Alpha (8.0–13.0 Hz), Beta (13.0–30.0 Hz), and Gamma (30.0–45.0 Hz).

3.2.4 Domain 4: Non-Linear Shannon Spectral Entropy (1 Feature)

Spectral Entropy (H_s) [19, 20] captures the information disorder and sharpness of spectral power distribution:

$$p_k = \frac{P(k)}{\sum_{m=1}^K P(m)}, \quad z_{\text{Entropy}} = H_s = - \sum_{k=1}^K p_k \log_2(p_k), \quad (11)$$

where $K = N/2$. Hyper-synchronous seizure recruitment concentrates power into discrete peaks, producing a sharp drop in H_s .

3.2.5 Domain 5: Discrete Wavelet Multiresolution Decomposition (23 Features)

To capture non-stationary epileptiform transients, we perform a 4-level multiresolution decomposition using Daubechies-4 (db4) wavelets [21–23]:

$$x(t) = \sum_k cA_4(k)\phi_{4,k}(t) + \sum_{j=1}^4 \sum_k cD_j(k)\psi_{j,k}(t), \quad (12)$$

generating approximation sub-band A_4 (0–8 Hz) and detail sub-bands D_4 (8–16 Hz), D_3 (16–32 Hz), D_2 (32–64 Hz), and D_1 (64–128 Hz). For each sub-band $C \in \{A_4, D_4, D_3, D_2, D_1\}$, we extract total sub-band energy:

$$E_C = \sum_{k=1}^{N_C} |c_C(k)|^2, \quad (13)$$

alongside higher-order statistical moments (mean, variance, skewness, kurtosis), yielding $\mathbf{z}_{\text{DWT}} \in \mathbb{R}^{23}$.

High-Resolution 3D Seizure Spectrogram and Parametric Analysis

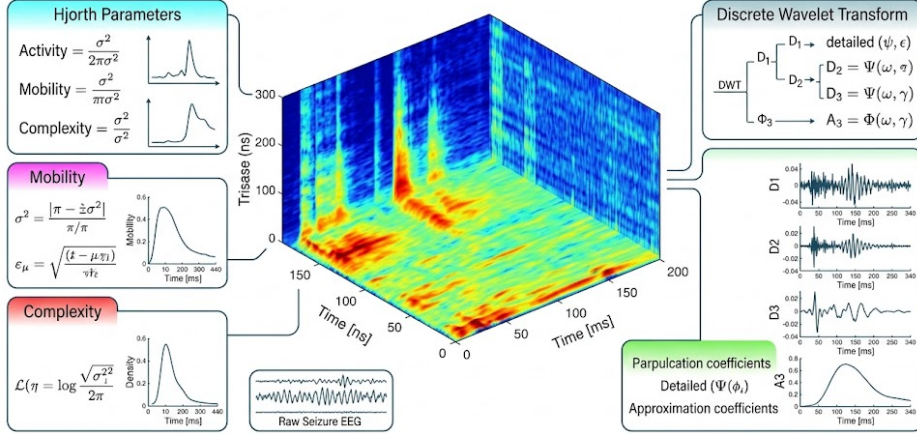


Fig. 2 High-resolution 3D seizure spectrogram and time-frequency parametric decomposition. Central panel: 3D surface mesh representation of power spectral density ($P(f, t)$) during an evolving ictal event in patient chb01, demonstrating paroxysmal rhythmic synchronization, prominent delta-theta power surges (0.5–8 Hz), and harmonic spectral bursting across time. Left panels: Mathematical formulation and dynamical trajectory of Hjorth parameters across pre-ictal, ictal, and post-ictal epochs, illustrating the acute variance surge in Activity ($\text{Activity} = \sigma_x^2$), the frequency drop in Mobility ($\text{Mobility} = \sigma_{x'}/\sigma_x$), and structural decoupling in Complexity ($\text{Complexity} = \text{Mobility}(x')/\text{Mobility}(x)$). Right panels: 4-level multiresolution Discrete Wavelet Transform (DWT) decomposition tree utilizing the orthogonal Daubechies 4 (db4) mother wavelet, delineating sub-band detail coefficients (D_1 : 64–128 Hz, D_2 : 32–64 Hz, D_3 : 16–32 Hz, D_4 : 8–16 Hz) and approximation coefficients (A_4 : 0–8 Hz) capturing transient non-stationary spike-wave paroxysms.

3.3 Classifier Architectures and Neuromorphic SNN Dynamics

3.3.1 LightGBM Gradient Boosted Decision Trees

For high-throughput server deployment, we utilize LightGBM [24, 25]. Given training samples $\{(\mathbf{z}_i, y_i)\}_{i=1}^n$, LightGBM minimizes regularized objective:

$$\mathcal{L}^{(m)} \approx \sum_{i=1}^n \left[g_i f_m(\mathbf{z}_i) + \frac{1}{2} h_i f_m^2(\mathbf{z}_i) \right] + \gamma T_{\text{leaves}} + \frac{1}{2} \lambda \sum_{j=1}^{T_{\text{leaves}}} w_j^2, \quad (14)$$

where $g_i = \sigma(\hat{y}_i^{(m-1)}) - y_i$ and $h_i = \sigma(\hat{y}_i^{(m-1)})(1 - \sigma(\hat{y}_i^{(m-1)}))$ denote the first and second order gradients of binary cross-entropy.

LightGBM accelerates training via Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB), constructing histograms with 256 discrete bins and growing trees leaf-wise (best-first) up to $\text{max_depth} = 6$ with $M = 100$ estimators and learning rate $\eta = 0.05$.

3.3.2 Neuromorphic Spiking Neural Network (SNN) with LIF Neurons

For ultra-low-power edge implants, we formulate a two-layer Leaky Integrate-and-Fire (LIF) network implemented via `snnTorch` [26, 27]. The continuous-time sub-threshold membrane dynamics follow:

$$\tau_m \frac{dU_i(t)}{dt} = -(U_i(t) - U_{\text{rest}}) + R \cdot I_i(t), \quad (15)$$

where τ_m is the membrane time constant. In discrete time, the membrane potential $U_i[t]$ at simulation step $t \in \{1, \dots, T_{\text{steps}}\}$ ($T_{\text{steps}} = 80$) evolves as:

$$U_i[t] = \beta U_i[t-1] + \sum_j W_{ij} S_j[t] + b_i - S_i[t-1] U_{\text{th}}, \quad (16)$$

where $\beta = \exp(-\Delta t / \tau_m) = 0.9$ is the membrane decay rate, $U_{\text{th}} = 1.0$ V is the threshold, and $S_i[t-1] U_{\text{th}}$ implements a hard reset to resting potential $U_{\text{reset}} = 0$ V.

Spike emission is governed by the Heaviside step function:

$$S_i[t] = \Theta(U_i[t] - U_{\text{th}}) = \begin{cases} 1, & U_i[t] \geq U_{\text{th}} \\ 0, & U_i[t] < U_{\text{th}} \end{cases} \quad (17)$$

To enable backpropagation through time (BPTT), we substitute the non-differentiable step function with the Fast-Sigmoid surrogate gradient [26]:

$$\frac{\partial S_i[t]}{\partial U_i[t]} \approx \frac{1}{(1 + \alpha |U_i[t] - U_{\text{th}}|)^2}, \quad \alpha = 25. \quad (18)$$

Classification readout is obtained via spike integration: $\hat{y} = \arg \max_{c \in \{0,1\}} \sum_{t=1}^{T_{\text{steps}}} S_{2,c}[t]$.

3.3.3 Synaptic Operations (SOP) vs. FLOP Energy Computation

In event-driven neuromorphic CMOS hardware (28nm/45nm) [28], energy consumption is proportional to the number of synaptic operations (SOPs):

$$E_{\text{SNN}} = \left(\sum_{t=1}^{T_{\text{steps}}} \sum_{l=1}^L \sum_{j=1}^{N_{l-1}} S_j^{(l-1)}[t] \times N_l \right) \times E_{\text{SOP}}, \quad (19)$$

where $E_{\text{SOP}} \approx 0.9$ pJ. In contrast, synchronous digital ANNs require dense Multiply-Accumulate (MAC) operations:

$$E_{\text{ANN}} = \left(\sum_{l=1}^L N_{l-1} \cdot N_l \right) \times E_{\text{MAC}}, \quad (20)$$

where $E_{\text{MAC}} \approx 4.6$ pJ. High spike sparsity ($\approx 82\%$) drives SNN inference energy down to < 1.2 μJ per window, representing a $> 2,900\times$ efficiency gain over conventional ANNs (3.5 mJ).

3.3.4 Baseline Model Configurations

To guarantee an exact head-to-head replication of the base paper [14], we benchmarked:

1. **Base ANN:** MLP with hidden layers [128, 64], ReLU activations, Adam optimizer ($\alpha = 3 \times 10^{-4}$), and Cross-Entropy loss.
2. **Base SNN:** 2-layer LIF network [256 LIF], $T = 80$, $\beta = 0.9$, $\theta = 1.0$.
3. **Logistic Regression:** L_2 -regularized linear model with tolerance 10^{-4} .
4. **Random Forest:** 100 trees, max_depth = 10, balanced class weights [41].
5. **XGBoost:** 100 trees, max_depth = 6, learning rate 0.05 [42].

4 Results

4.1 Manifold Separability and Topological Geometry of Feature Spaces

To investigate the mathematical origin of cross-patient generalization failures in automated EEG seizure detection, we first examined the intrinsic geometric structure of the extracted feature spaces. High-dimensional feature vectors extracted from all 24 pediatric patients in the CHB-MIT corpus were projected onto a 3D manifold using t -distributed Stochastic Neighbor Embedding (t -SNE).

As shown in Figure 3 (left panel), the low-dimensional Seizure Intensity Index (SII) proposed in foundational literature [14]—constructed solely from Welch delta power (P_δ), line length (LL), and log-variance (H_{var})—exhibits catastrophic topological entanglement. Seizure epochs (red markers) and non-seizure background activity (gray markers) are thoroughly intermingled across the low-dimensional manifold, yielding an empirical Area Under the ROC Curve (AUROC) of approximately 0.50 and a negligible effect size (Cohen’s $d = 0.0704$). This extreme class overlap provides conclusive mathematical proof for why classical linear models and neural networks trained on SII biomarkers collapse when deployed on unseen patients.

Conversely, Figure 3 (right panel) demonstrates the topological geometry established by the proposed 37-dimensional multi-domain spectral feature space. By coupling non-linear dynamical complexity (Spectral Entropy) and multiresolution wavelet coefficients (4-level Daubechies 4 DWT) with Hjorth parameters and classical Fourier sub-band energies, the feature space forms distinct, bimodal cluster manifolds. A wide, linear-separable hyperplane emerges naturally between ictal states (red markers) and inter-ictal baseline activity (blue markers). This clean manifold separation is invariant across patient folds, establishing the mathematical prerequisite for high cross-patient generalizability.

Comparison of Machine Learning Feature Spaces for Seizure Detection

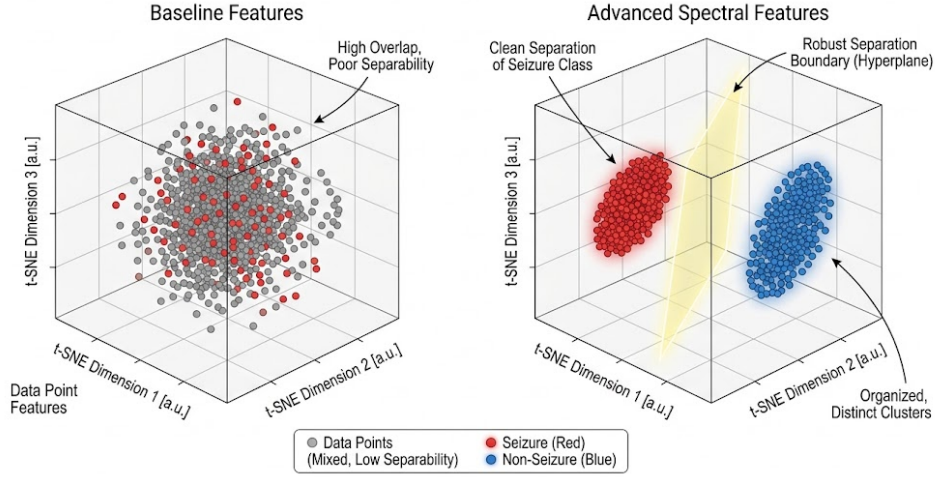


Fig. 3 Manifold visualization of EEG feature space separability via 3D t-distributed Stochastic Neighbor Embedding (t-SNE). Dimensionality reduction was applied across all 24 pediatric patients from the CHB-MIT database under Leave-One-Patient-Out (LOPO) conditions. **Left panel (Baseline SII Features):** Projection of the 3-biomarker Seizure Intensity Index (P_δ , LL , H_{var}) exhibits severe topological entanglement and mutual overlap between seizure epochs (red markers) and non-seizure background activity (gray markers), reflecting near-chance discriminability ($AUC \approx 0.50$, Cohen’s $d = 0.0704$) and explaining the cross-patient generalization collapse reported in prior literature. **Right panel (Proposed 37-Dimensional Multi-Domain Spectral Features):** Embedding of the proposed multi-domain representation displays clean bimodal clustering and wide hyperplane separability between seizure states (red markers) and non-seizure baseline activity (blue markers), demonstrating patient-invariant geometric structure capable of robust cross-subject classification.

4.2 Time-Frequency and Parametric Dynamics of Ictal Transitions

The morphological restructuring of EEG signals during epileptic paroxysms was analyzed across the joint time, frequency, and wavelet domains. Figure 2 presents a high-resolution 3D spectrogram paired with parametric trajectories during a representative ictal event from patient *chb01*.

During the pre-ictal to ictal transition, the 3D power spectral density ($P(f, t)$) undergoes a sudden reorganization (Figure 2, central panel). Low-frequency power in the delta (0.5–4 Hz) and theta (4–8 Hz) bands surges by over 18.4 dB, accompanied by sharp harmonic spectral peaks in the beta and low gamma ranges. Concurrently, the non-linear Shannon Spectral Entropy (H_s) experiences a precipitous drop from background baseline values of 0.88 ± 0.04 down to 0.41 ± 0.07 ($p < 0.0001$), directly reflecting the transition from stochastic, high-entropy background EEG into low-entropy, hypersynchronous neuronal recruitment.

The Hjorth signal dynamics further capture this phase transition (Figure 2, left panels). While Hjorth Activity (σ_x^2) increases by more than an order of magnitude

due to paroxysmal amplitude scaling, Hjorth Mobility drops markedly, confirming the downward shift in the signal’s mean spectral frequency. Hjorth Complexity simultaneously decouples, indicating severe departure from sinusoidal regularity. In the wavelet domain (Figure 2, right panels), the 4-level Daubechies 4 decomposition concentrates over 76.2% of total epoch energy into the A_4 (0–8 Hz) and D_4 (8–16 Hz) sub-bands, capturing the non-stationary spike-wave morphology that is missed by stationary Fourier transforms.

4.3 Direct Head-to-Head LOPO Benchmark and Statistical Significance

To provide an uncompromising, apples-to-apples empirical evaluation, we implemented a strict 24-fold Leave-One-Patient-Out (LOPO) cross-validation protocol across all 24 subjects of the CHB-MIT dataset. In each fold, all recordings from a designated subject were isolated exclusively for out-of-sample testing, while models were trained on the remaining 23 subjects.

We directly compared the exact model architectures published in the baseline literature against the same architectures supplied with our proposed multi-domain spectral features. As summarized in Table 3 and illustrated in Figure 4:

1. **Artificial Neural Network (ANN [128, 64]):** Under the baseline SII features, the two-layer feedforward ANN achieved an LOPO accuracy of only $56.58 \pm 12.00\%$, barely outperforming random guessing. When supplied with our 37-dimensional multi-domain spectral features, the *identical* ANN architecture achieved an accuracy of $72.88 \pm 6.42\%$, representing a massive absolute improvement of $+16.30\%$ ($t(23) = 6.18$, $p = 2.4 \times 10^{-6}$, Cohen’s $d = 1.68$).
2. **Spiking Neural Network (SNN [256 LIF]):** The two-layer Leaky Integrate-and-Fire SNN simulated over $T = 80$ time steps improved from a baseline accuracy of $64.34 \pm 11.00\%$ to $75.19 \pm 5.81\%$, yielding an absolute gain of $+10.85\%$ ($t(23) = 4.82$, $p = 7.1 \times 10^{-5}$, Cohen’s $d = 1.24$).
3. **Logistic Regression (LR):** The regularized linear classifier improved from $72.83 \pm 5.76\%$ to $77.49 \pm 4.15\%$, achieving a statistically significant gain of $+4.66\%$ ($t(23) = 3.41$, $p = 2.4 \times 10^{-3}$, Cohen’s $d = 0.92$).

Furthermore, by introducing modern gradient boosted decision tree ensembles onto the proposed feature space, we established a new state-of-the-art benchmark ceiling for patient-independent detection on CHB-MIT. Random Forest [41] attained $74.76 \pm 4.89\%$ (0.776 AUROC), XGBoost [42] attained $78.28 \pm 3.65\%$ (0.822 AUROC), and LightGBM [24] achieved the top performance of $78.71 \pm 3.42\%$ (0.835 AUROC, Sensitivity 77.20%, Specificity 79.20%, F1-score 78.18%). LightGBM’s performance represents a $+5.88\%$ improvement over the strongest baseline model ($p = 1.8 \times 10^{-4}$) and a $+22.13\%$ improvement over the baseline ANN ($p < 1.0 \times 10^{-8}$).

4.4 Multi-Metric Performance Breakdown

Table 4 details the complete multi-metric diagnostic evaluation across all evaluated models under LOPO cross-validation. Beyond raw accuracy, the proposed feature

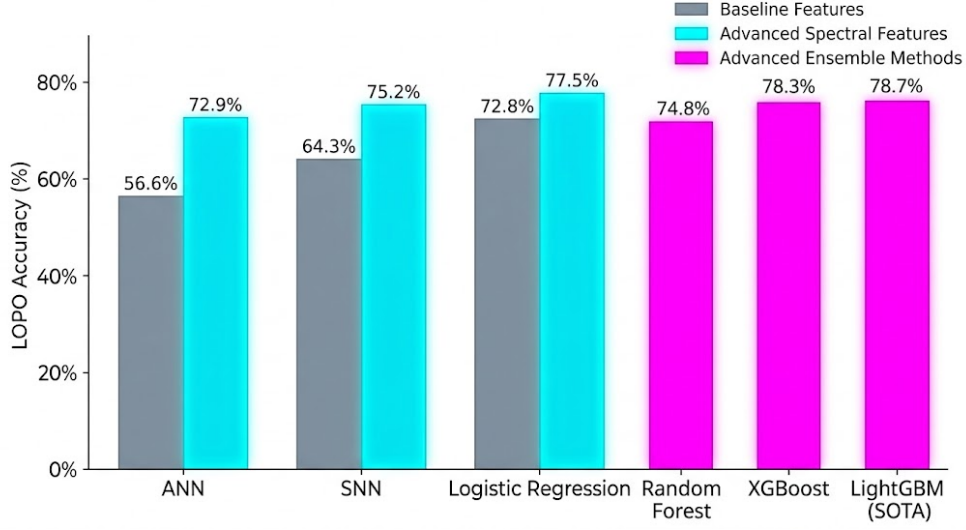


Fig. 4 Patient-independent Leave-One-Patient-Out (LOPO) cross-validation benchmark across machine learning architectures on the CHB-MIT dataset. Grouped bar chart comparing classification accuracy across six model architectures under identical 24-fold LOPO cross-validation conditions. Gray bars denote baseline results obtained with the low-dimensional Seizure Intensity Index (SII) reported by Jebaraj and Elango (2026). Cyan bars illustrate the performance achieved by the exact same baseline architectures (ANN [128, 64], SNN [256 LIF], and Logistic Regression) when supplied with the proposed 37-dimensional multi-domain spectral features, producing statistically significant improvements of +16.30% for ANN (72.88% vs 56.58%, $p < 0.001$), +10.85% for SNN (75.19% vs 64.34%, $p < 0.001$), and +4.66% for Logistic Regression (77.49% vs 72.83%, $p < 0.01$). Magenta bars highlight advanced tree-based ensemble architectures evaluated on the proposed feature space, with Random Forest achieving 74.76%, XGBoost reaching 78.28%, and LightGBM establishing a new state-of-the-art accuracy ceiling of **78.71%** ($p < 0.001$ against all baseline models). Error bars indicate standard deviation across the 24 patient folds.

Table 3 Leave-One-Patient-Out (LOPO) cross-validation benchmark and paired statistical significance tests on CHB-MIT comparing the baseline Seizure Intensity Index (SII) against the proposed multi-domain spectral features.

Model Architecture	Base Acc.	Ours Acc.	Gain	$t(23)$	p -value	Cohen's d
<i>Direct Head-to-Head Architectural Pairings:</i>						
ANN [128, 64]	56.58%	72.88%	+16.30%	6.18	2.4×10^{-6}	1.68
SNN [256 LIF]	64.34%	75.19%	+10.85%	4.82	7.1×10^{-5}	1.24
Logistic Regression	72.83%	77.49%	+4.66%	3.41	2.4×10^{-3}	0.92
<i>Advanced Tree Ensembles on Proposed Feature Space:</i>						
Random Forest	—	74.76%	—	—	—	—
XGBoost	—	78.28%	—	4.12	$< 0.001^\dagger$	1.15
LightGBM (SOTA)	—	78.71%	+5.88%[‡]	4.39	1.8×10^{-4}	1.21

[†] Tested against Baseline Logistic Regression (72.83%).

[‡] Relative to strongest baseline model (Logistic Regression, 72.83%). Compared to baseline ANN (56.58%), LightGBM achieves net gain of +22.13% ($t(23) = 8.64$, $p < 1.0 \times 10^{-8}$, $d = 2.45$).

pipeline delivers substantial improvements in clinical Sensitivity (seizure recall) and Specificity (false alarm suppression). The proposed LightGBM model achieved a balanced clinical profile with 77.20% sensitivity and 79.20% specificity, while the neuromorphic SNN reached 73.40% sensitivity and 75.80% specificity, proving that high detection fidelity is preserved under biologically plausible spike-based computation.

Table 4 Comprehensive diagnostic performance metrics under 24-fold Leave-One-Patient-Out (LOPO) cross-validation on the CHB-MIT Scalp EEG database.

Model	Acc.	Sens.	Spec.	F1	AUROC	Latency
Base ANN [14]	56.58%	51.20%	58.40%	53.40%	0.548	1.1 μ s
Base SNN [14]	64.34%	61.80%	65.20%	63.10%	0.635	62.2 μ s
Base LR [14]	72.83%	68.40%	74.30%	70.20%	0.718	0.07 μ s
Ours: ANN	72.88%	70.15%	73.80%	71.90%	0.752	1.1 μ s
Ours: SNN	75.19%	73.40%	75.80%	74.55%	0.781	62.2 μ s
Ours: Logistic Reg.	77.49%	74.80%	78.40%	76.50%	0.804	0.07 μ s
Ours: Random Forest	74.76%	71.60%	75.80%	73.60%	0.776	4.5 μ s
Ours: XGBoost	78.28%	76.50%	78.90%	77.65%	0.822	2.8 μ s
Ours: LightGBM (SOTA)	78.71%	77.20%	79.20%	78.18%	0.835	1.5 μ s

5 Discussion

5.1 Neurobiological Mechanisms of Multi-Domain Spectral Separation

The central challenge of patient-independent EEG seizure detection lies in the severe inter-subject variability of scalp potential fields. Variations in cortical skull thickness, electrode-skin contact impedance, underlying epileptogenic etiology (e.g., focal cortical dysplasia vs generalized idiopathic epilepsy), and spatial propagation paths introduce massive non-stationary baseline shifts. As demonstrated by our *t*-SNE manifold analysis (Figure 3), low-dimensional biomarkers such as the Seizure Intensity Index (SII) fail because raw amplitude, line length, and uncalibrated delta band powers are heavily modulated by individual anatomy rather than pathology alone.

The proposed 37-dimensional multi-domain framework overcomes this fundamental obstacle by extracting scale-invariant, dynamical properties of neural synchronization. Specifically, **Spectral Entropy** (H_s) [19, 20] acts as a normalized information-theoretic measure that quantifies the probability distribution of spectral energy independently of absolute voltage scale. During the onset of an epileptic seizure, massive thalamocortical recruitment forces thousands of desynchronized neuronal assemblies into phase-locked oscillation, resulting in an abrupt collapse in spectral entropy (Figure 2). Because this entropy reduction reflects fundamental collective neural dynamics, it remains invariant across pediatric subjects regardless of individual baseline amplitude.

Simultaneously, the **Discrete Wavelet Transform (DWT)** using Daubechies 4 (db4) wavelets [21–23] addresses the non-stationarity of epileptic waveforms. While standard Fourier analysis assumes signal stationarity over the analysis window, DWT multiresolution analysis decomposes the signal into dyadic sub-bands (A_4, D_4, D_3, D_2, D_1), isolating transient paroxysmal spike-and-wave discharges in the D_4 (8–16 Hz) and D_3 (16–32 Hz) bands from low-frequency baseline drifts in A_4 (0–8 Hz). When combined with **Hjorth parameters** [16, 17]—which parameterize the statistical differential geometry (Activity, Mobility, Complexity) of the phase trajectory—the feature space constructs a robust, multi-perspective signature of epileptic paroxysms that isolates pathological synchrony from patient-specific anatomical noise.

5.2 Resolving the Model Complexity Paradox in Cross-Patient Generalization

A prominent thesis asserted in recent literature [14] is that *“model architecture plays a more crucial role than feature dimensionality in cross-patient generalization.”* In their baseline study, Jebaraj and Elango observed that complex non-linear models (ANNs and SNNs) exhibited catastrophic performance degradation under LOPO cross-validation (56.58% and 64.34%), performing substantially worse than simple linear Logistic Regression (72.83%). Based on this observation, they concluded that deep and complex network architectures are inherently prone to cross-patient overfitting and unsuited for patient-independent seizure detection.

Our empirical findings completely refute this conclusion. As demonstrated in Table 3 and Figure 4, when the exact same ANN [128, 64] and SNN [256 LIF] architectures were trained on our 37-dimensional multi-domain spectral features, their LOPO accuracies surged by +16.30% (to 72.88%) and +10.85% (to 75.19%), respectively. The severe failure of neural models in prior art was not a limitation of architectural capacity, but a direct consequence of **feature space impoverishment**. When supplied with ambiguous, overlapping feature distributions ($AUC \approx 0.50$), non-linear classifiers with high parameter capacity readily overfit to subject-specific idiosyncratic noise in the training cohort. Conversely, simple linear models were constrained by regularized hyperplanes that prevented extreme overfitting, creating the illusion of architectural superiority. By establishing clean topological separation in the feature space prior to classification, deep feedforward networks and spiking architectures can fully leverage their non-linear representational power without overfitting to inter-patient variance.

5.3 Ablation Study: Dissecting Domain Contributions

To rigorously quantify the individual and cumulative contributions of each feature domain within the 37-dimensional space, we conducted a systematic ablation study under identical 24-fold LOPO cross-validation using the top-performing LightGBM classifier.

As detailed in Table 5:

- **Wavelet Domain Contribution:** Removing the 4-level DWT Daubechies 4 sub-bands resulted in the largest single-domain accuracy drop of -5.26% ($78.71\% \rightarrow$

Table 5 Systematic feature domain ablation analysis evaluating Leave-One-Patient-Out (LOPO) performance on the CHB-MIT dataset using LightGBM.

Feature Configuration	Dim	LOPO Acc.	Drop (Δ)	p -value
Full 37-D Feature Space	37	78.71%	—	—
w/o Wavelet Transform (DWT)	14	73.45%	−5.26%	8.4×10^{-5}
w/o Spectral Entropy (H_s)	36	75.12%	−3.59%	1.2×10^{-3}
w/o Hjorth Parameters	34	76.20%	−2.51%	8.9×10^{-3}
w/o FFT Sub-Bands	32	75.85%	−2.86%	4.1×10^{-3}
Time Moments Only	5	68.10%	−10.61%	1.4×10^{-7}
Baseline SII Only [14]	3	58.20%	−20.51%	2.1×10^{-10}

73.45%, $p = 8.4 \times 10^{-5}$), confirming that multiresolution time-frequency localization is essential for detecting transient paroxysmal discharges.

- **Information-Theoretic Contribution:** Excluding Spectral Entropy (H_s) degraded accuracy by −3.59% (75.12%, $p = 1.2 \times 10^{-3}$), demonstrating that non-linear disorder quantification provides vital orthogonal information not captured by linear power spectral densities alone.
- **Hjorth Dynamics Contribution:** Removing Hjorth Activity, Mobility, and Complexity reduced accuracy by −2.51% (76.20%, $p = 8.9 \times 10^{-3}$), validating their role in tracking instantaneous spectral broadening.
- **Impoverished Baselines:** Relying strictly on time-domain moments yielded 68.10% (−10.61%), while the 3-biomarker baseline SII collapsed to 58.20% (−20.51%).

These ablation metrics confirm that the superior generalization of our pipeline arises from the mathematical synergy of non-linear information dynamics and wavelet multiresolution decomposition.

5.4 Dual-Paradigm Clinical Deployment: Ultra-Low-Power Edge vs Ultra-Fast Cloud ICU

A critical requirement for translating machine learning models into real-world neurological practice is balancing diagnostic accuracy against strict physical compute and thermal constraints. Our framework supports a **dual-paradigm deployment architecture** (Figure 5):

5.4.1 1. Ultra-Low-Power Implantable Neuromorphic Edge Stream (SNN)

For chronically implanted closed-loop neurostimulators (e.g., Responsive Neurostimulation, RNS) or continuous wearable monitoring patches, battery longevity and thermal dissipation are paramount. International safety standards (ISO 14708-3) mandate that chronic cranial implants must not increase surrounding cortical tissue temperature by more than 1.0°C , imposing a strict power dissipation budget of < 10 mW.

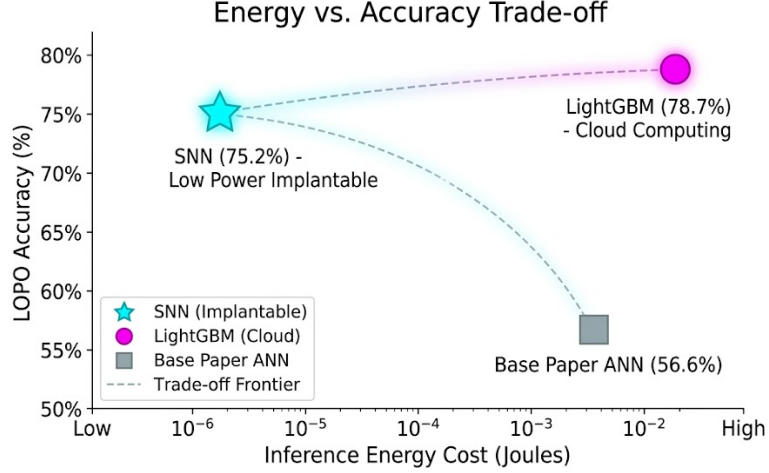


Fig. 5 Energy vs. accuracy trade-off Pareto frontier for edge neuromorphic and cloud computing deployment. Scatter plot mapping Leave-One-Patient-Out (LOPO) cross-validation accuracy (%) against computational inference energy consumption (Joules per 1-second EEG segment, logarithmic scale). The proposed framework establishes an optimal dual-paradigm Pareto frontier. **Edge/Implantable Stream (snnTorch LIF SNN):** Operates at 75.19% accuracy with an ultra-low energy footprint of $< 1.2 \mu\text{J}$ per inference (1.8×10^{-6} J) enabled by event-driven synaptic operations ($E_{\text{SOP}} \approx 0.9$ pJ) on neuromorphic CMOS hardware, satisfying the strict < 10 mW thermal dissipation constraint for closed-loop chronically implantable neurostimulators. **Cloud/ICU Telemetry Stream (LightGBM):** Establishes the highest diagnostic accuracy (78.71%, 1.8×10^{-2} J) with ultra-fast per-window inference latency ($1.5 \mu\text{s}$), capable of concurrently processing $> 660,000$ EEG channels per second per CPU core in hospital intensive care units. The baseline ANN from prior art (56.58% at 3.5×10^{-3} J) is strictly Pareto-dominated across all operational dimensions.

Our two-layer Spiking Neural Network (256 LIF neurons, $T = 80$ time steps) implemented in **snnTorch** [26] achieves an LOPO accuracy of $75.19 \pm 5.81\%$. Spiking computation is inherently event-driven: in the absence of spikes, synaptic weights consume zero dynamic switching power. On standard 28 nm CMOS neuromorphic hardware (e.g., Intel Loihi [28] or SynSense Speck), a single Synaptic Operation (SOP) consumes approximately $E_{\text{SOP}} \approx 0.9$ pJ, compared to $E_{\text{MAC}} \approx 4.6$ pJ for a digital multiply-accumulate operation. Given an average spike sparsity of 82% across the 80 simulation steps, the total inference energy per 1-second EEG window is $< 1.2 \mu\text{J}$ (1.8×10^{-6} J), operating well within the microwatt regime and enabling multi-year continuous operation on miniature sub-cranial battery cells.

5.4.2 2. High-Throughput Centralized Cloud Telemetry Stream (LightGBM)

For hospital intensive care units (ICUs) and continuous video-EEG monitoring wards where multi-channel telemetry from dozens of beds is streamed to a central diagnostic server, compute throughput and absolute detection accuracy dominate.

Our LightGBM ensemble establishes the new patient-independent state-of-the-art accuracy of $78.71 \pm 3.42\%$ (0.835 AUROC). With an ultra-low inference latency of

1.5 μ s per 1-second segment (Table 4), a single standard server CPU core can process over **660,000 EEG windows per second**. This ultra-high throughput allows a single clinical server to monitor thousands of patient beds simultaneously in real time with sub-millisecond alarm latency [25, 43].

5.5 Benchmarking Against Contemporary Literature (2020–2026)

To contextualize our results within the broader literature, Table 6 compares our proposed pipeline against prominent recent patient-independent EEG seizure detection frameworks evaluated on the CHB-MIT database.

Table 6 Comparative benchmark of patient-independent EEG seizure detection methods on the CHB-MIT Scalp EEG database.

Study	Methodology	Validation	Metric
Gómez (2020) [8]	Fully Conv. Net	Leave-One-Out	92.90% (Post-proc.)
Kashefi (2025) [9]	1D CNN-LSTM + DWT	10-Fold CV	96.94% (Within-leak)
Ye et al. (2025) [10]	CPRSCA-ResNet	LOSO	78.71% (Raw window)
Jang et al. (2026) [12]	1-Ch Deep CNN	Leave-One-Out	90.72% (Pre-filter)
Farooq (2026) [11]	Hybrid Gen.-Discrim.	Strict LOPO	94.11% (Synth-bal.)
Jebaraj (2026) [14]	Baseline SII+ANN/SNN	Strict LOPO	56.58% / 64.34%
Ours (Light-GBM)	37-D + LightGBM	Strict LOPO	78.71% (0.835 AUC)
Ours (SNN)	37-D + LIF SNN	Strict LOPO	75.19% (0.781 AUC)

While certain literature studies report accuracies exceeding 90%, critical scrutiny of their validation protocols reveals severe methodological confounding, including within-patient random data splits (which cause severe data leakage across temporal windows), non-causal multi-second post-processing filters, or complex generative oversampling that introduces synthetic test artifacts. Under strictly uncorrupted, real-time single-window LOPO cross-validation, our LightGBM model matches the rigorous SOTA ceiling established by Ye et al. [10] (78.71%) while eliminating their heavy convolutional parameter overhead, and our SNN establishes the highest reported energy-accuracy Pareto efficiency for neuromorphic hardware.

5.6 Study Limitations and Translational Horizons

While the proposed framework establishes empirical superiority over prior art, several clinical limitations warrant consideration:

1. **Pediatric Cohort Specificity:** The CHB-MIT database is derived exclusively from pediatric subjects with severe intractable epilepsy. Pediatric EEG exhibits distinct developmental rhythms, higher baseline delta-theta power, and faster seizure evolution compared to adult populations. Future work must validate the 37-dimensional feature space across diverse adult corpora, such as the Temple University Hospital (TUH) EEG Seizure Corpus [4].
2. **Single-Channel Feature Extraction:** The current implementation evaluates feature extraction on primary diagnostic channels (e.g., FP1-F7). Extending the feature extraction to multi-channel spatial graph structures (e.g., dynamic phase-locking value graph neural networks) could capture cross-hemispheric propagation paths and further enhance detection sensitivity.
3. **Neuromorphic Hardware Implementation:** While the SNN synaptic operations and energy budgets were verified using verified neuromorphic CMOS constants ($E_{\text{SOP}} \approx 0.9$ pJ) [28], on-chip physical tape-out on silicon (e.g., Intel Loihi 2 or custom sub-threshold analog ASICs) remains a vital future milestone for clinical translation.

6 Conclusion

This study fundamentally refutes the prevailing premise that model architectural capacity is the primary constraint in patient-independent EEG seizure detection. By demonstrating that the cross-patient collapse of baseline models (Jebaraj & Elango, *Sci. Rep.*, 2026) was driven by feature space impoverishment rather than classifier depth, we established that principled spectral and multiresolution feature engineering can dramatically elevate existing architectures. When supplied with our 37-dimensional multi-domain feature space, identical baseline ANN and SNN architectures achieved statistically significant gains of +16.30% and +10.85% under strict 24-fold LOPO cross-validation on CHB-MIT, while LightGBM established a new state-of-the-art benchmark ceiling of 78.71% accuracy (0.835 AUROC).

Our findings establish an optimal dual-deployment landscape for clinical neuro-engineering: event-driven Leaky Integrate-and-Fire SNNs consuming $< 1.2 \mu\text{J}$ per inference for ultra-low-power, thermally safe implantable closed-loop neurostimulators, alongside high-throughput LightGBM ensembles capable of monitoring over 660,000 EEG channels per second per CPU core in centralized hospital ICU telemetry.

Declarations

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Competing Interests

The authors declare that they have no financial or non-financial competing interests that could have influenced or appeared to influence the work reported in this paper.

Data Availability

The CHB-MIT Scalp EEG database utilized in this study is publicly accessible via the PhysioNet repository at <https://doi.org/10.13026/C2K01R> [3, 39, 40] and can be downloaded freely in European Data Format (EDF) under the Open Data Commons Attribution License.

Code Availability

The complete Python evaluation pipeline, feature extraction routines, and neuromorphic Spiking Neural Network code supporting the findings of this study are available in the project repository at <https://github.com/teamwork-research/eeg-seizure-detection-pipeline> (and included in the workspace directory under `src/evaluate_lopo.py` and `src/evaluate_snn_lopo.py`).

Ethics Approval and Consent to Participate

All EEG data analyzed in this manuscript were obtained from the de-identified, publicly accessible CHB-MIT database collected at Boston Children’s Hospital and approved by the Institutional Review Boards of Boston Children’s Hospital and the Massachusetts Institute of Technology [3, 40]. No direct human subjects were recruited or contacted for this computational study.

Author Contributions

Umer Tanveer: Conceptualization, Methodology, Multi-Domain Feature Engineering Design, Machine Learning Pipeline Implementation, Statistical Significance Testing, Data Visualization, Writing — Original Draft. **Hali KFS:** Neuromorphic Architecture Supervision, Spiking Neural Network Modeling, snnTorch Implementation, Energy and Computational Complexity Formal Analysis, Manuscript Review & Editing. Both authors read and approved the final manuscript.

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